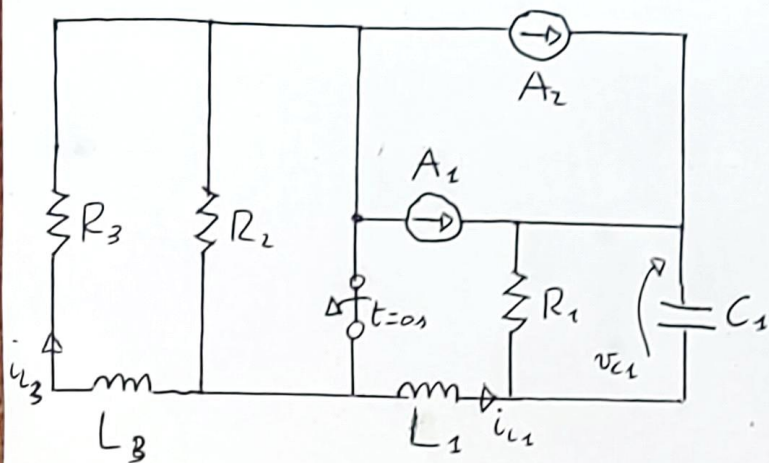


TRANSITORIO - METODO PER ISPEZIONE



$$\begin{aligned} R_1 &= 1 \, \Omega & C_1 &= 1 \, \text{mF} \\ R_2 &= 4 \, \Omega & A_1 &= 3 \, \text{A} \\ R_3 &= 2 \, \Omega & A_2 &= 4 \, \text{A} \\ L_1 &= 3 \, \text{mH} \\ L_3 &= 1 \, \text{mH} \end{aligned}$$

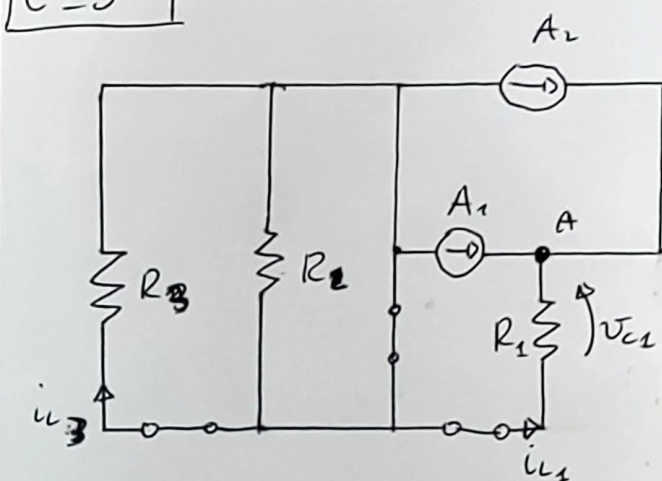
DETERMINARE:

$$t = 0^- : W_{L1}, W_{L3}, W_{C1}$$

$$t = 0^+ : \frac{di_{L3}}{dt}, \frac{dv_{C1}}{dt}$$

$$t = \infty : P_{R2}, Q_{C1}, P_{A2}$$

$t = 0^-$



$$LUC(A) : A_1 + A_2 + i_{L1} = 0$$

$$i_{L1} = -A_1 - A_2 = -7 \, \text{A}$$

$W_{L1} = \frac{1}{2} L_1 i_{L1}^2 = 73.5 \, \text{mJ}$

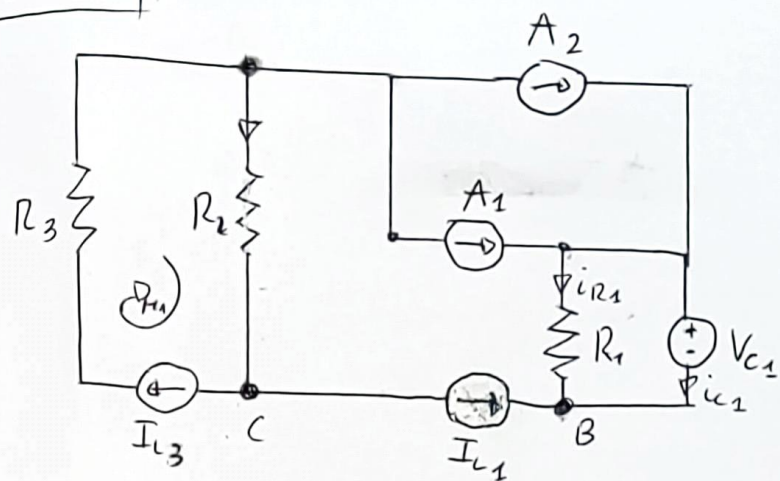
$$v_{C1} = -v_{R1} = -i_{L1} R_1 = 7 \, \text{V}$$

$W_{C1} = \frac{1}{2} C_1 v_{C1}^2 = 24.5 \, \text{mJ}$

RAMI CON R_2 e R_3 SONO IN CORTO

$$i_{L3} = 0 \, \text{A} \quad \rightarrow \quad \boxed{W_{L3} = 0 \, \text{J}}$$

$t = 0^+$



$$V_{c1} = 7V, I_{L1} = -7A, I_{L3} = 0A$$

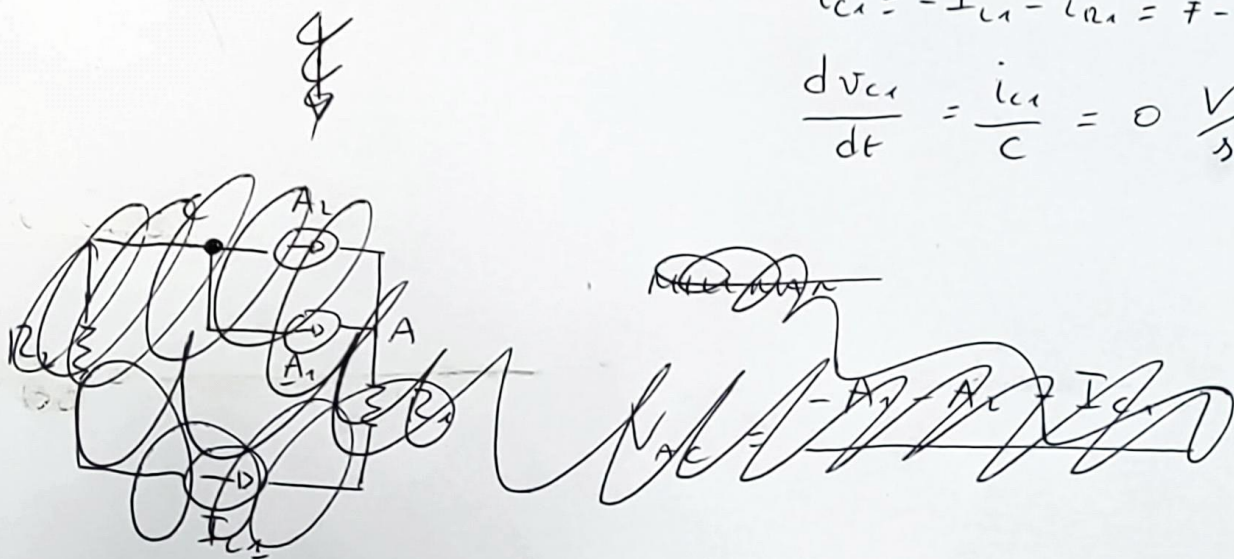
$$V_{c1} = v_{R1} = 7V$$

$$i_{R1} = \frac{v_{R1}}{R1} = 7A$$

$$KCL(B): I_{L1} + i_{R1} + i_{c1} = 0$$

$$i_{c1} = -I_{L1} - i_{R1} = 7 - 7 = 0A$$

$$\frac{dv_{c1}}{dt} = \frac{i_{c1}}{C} = 0 \frac{V}{s}$$



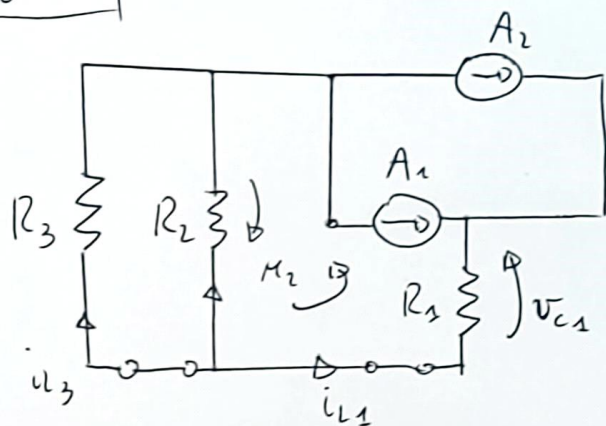
$$I_{L3} = 0A \rightarrow v_{R3} = I_{L3} R3 = 0V$$

$$KVL(M1): -v_{L3} - v_{R3} = v_{R2} \rightarrow v_{L3} = -v_{R2} = -I_{L1} R2 = 28V$$

$$KCL(C): I_{L3} + I_{L1} - i_{R2} = 0 \rightarrow I_{L1} = i_{R2}$$

$$\frac{di_{L3}}{dt} = \frac{v_{L3}}{L3} = 28000 A/s$$

$$t = \infty$$



$$P_{R_2} = R_2 \left((A_1 + A_2) \frac{R_3}{R_2 + R_3} \right)^2 =$$

$$= 21.7 \text{ W}$$

$$Q_{C_1} = V_{C_1} C_1 = R_1 (A_1 + A_2) C_1 =$$

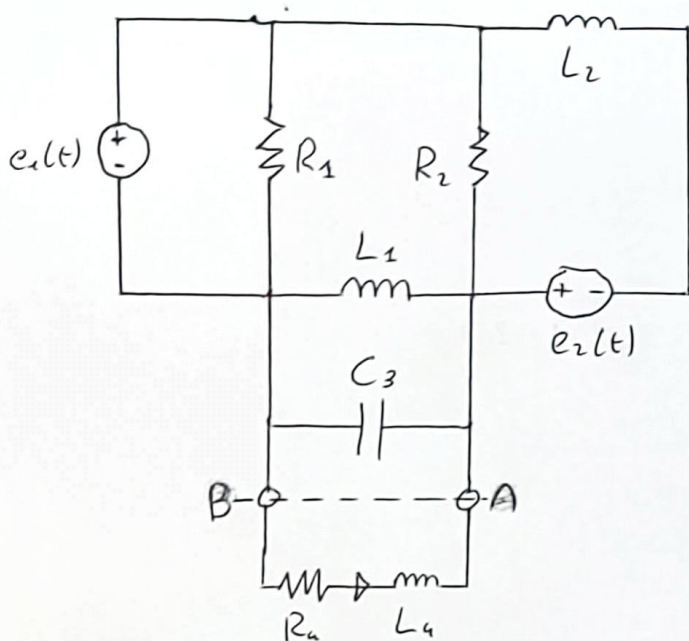
$$= 7 \text{ mC}$$

$$\text{LKT}(M_2) : -V_{A_2} + V_{C_1} + V_{R_2} = 0$$

$$V_{A_2} = +V_{C_1} + R_2 \left((A_1 + A_2) \frac{R_3}{R_2 + R_3} \right) = +7 + 9.3 = +16.3 \text{ V}$$

$$P_{A_2} = V_{A_2} A_2 = 65.3 \text{ W}$$

REGIME SINUSOIALE



$$e_1(t) = 3 \cos(2t + 75^\circ) \text{ V}$$

$$e_2(t) = \sin(2t - 165^\circ) \text{ V} \\ = \cos(2t - 255^\circ) \text{ V}$$

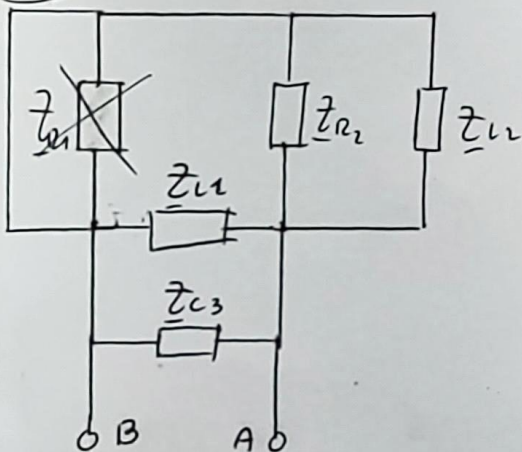
$$R_1 = 1 \Omega, R_2 = 3 \Omega, L_1 = 3 \text{ mH}, \\ L_2 = 1 \text{ mH}, C_3 = 3 \text{ mF}, R_a = 5 \Omega, \\ L_4 = 2 \text{ mH}$$

$$\underline{Z}_{R_1} = 1 \Omega; \underline{Z}_{R_2} = 3 \Omega; \underline{Z}_{L_1} = j\omega L_1 = 6 \text{ j m}\Omega; \underline{Z}_{L_2} = 2 \text{ j m}\Omega;$$

$$\underline{Z}_{C_3} = \frac{1}{j\omega C_3} = -166.7 \text{ j}\Omega; \underline{Z}_{R_a} = 5 \Omega, \underline{Z}_{L_4} = 4 \text{ j m}\Omega.$$

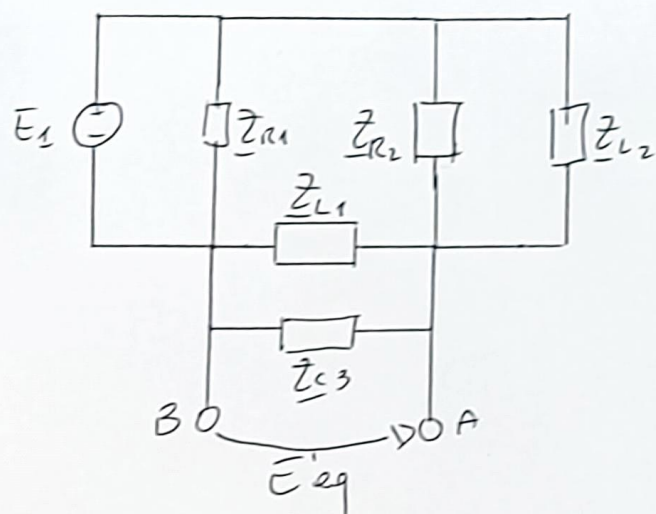
$$\underline{E}_1 = \frac{3}{\sqrt{2}} \angle 75^\circ \text{ V} \quad \underline{E}_2 = \frac{1}{\sqrt{2}} \angle -255^\circ \text{ V}$$

$\underline{Z}_{eq}$



$$\underline{Z}_{eq} = \left(\left(\underline{Z}_{R_2} \parallel \underline{Z}_{L_1} \right) \parallel \underline{Z}_{L_2} \right) \parallel \underline{Z}_{C_3} \\ = 0 + 0.0015 \text{ j}\Omega$$

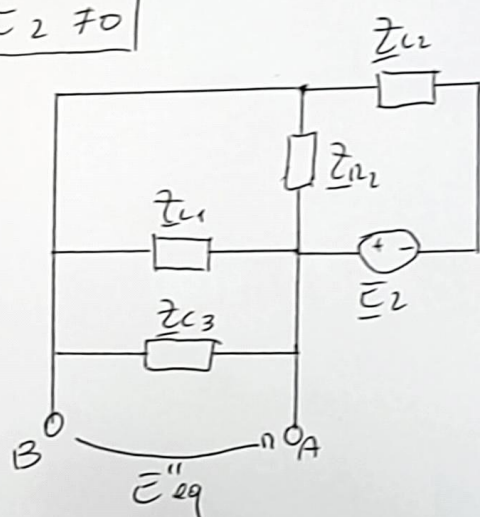
$$\underline{E}_1 \neq 0$$



$$\underline{E}'_{eq} = \underline{E}_1 \frac{(\underline{z}_{L1} \parallel \underline{z}_{C3})}{(\underline{z}_{L1} \parallel \underline{z}_{C3}) + (\underline{z}_{R1} \parallel \underline{z}_{L2})}$$

$$= 0.41 + j1.54 \text{ V} = 1.59 \angle 75^\circ \text{ V}$$

$$\underline{E}_2 \neq 0$$

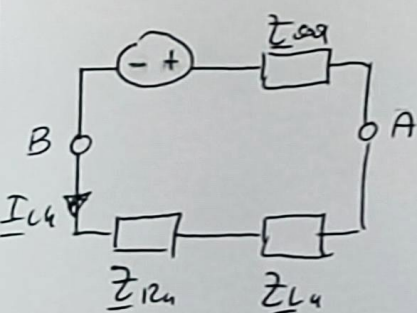


$$\underline{E}''_{eq} = \frac{(\underline{z}_{C3} \parallel \underline{z}_{L1} \parallel \underline{z}_{R1})}{(\underline{z}_{C3} \parallel \underline{z}_{L1} \parallel \underline{z}_{R1}) + \underline{z}_{L2}} \cdot \underline{E}_2$$

$$= -0.14 + j0.51 \text{ V}$$

$$= 0.53 \angle 105^\circ \text{ V}$$

$$\underline{E}_{eq} = \underline{E}'_{eq} + \underline{E}''_{eq} = 0.27 + j2.05 \text{ V} = 2.07 \angle 82^\circ \text{ V}$$



$$\underline{I}_{Lu} = - \frac{\underline{E}_{eq}}{\underline{z}_{eq} + \underline{z}_{Ru} + \underline{z}_{Lu}} = 0.41 \angle 97.7^\circ \text{ A}$$

$$i_{Lu}(t) = 0.58 \cos(2t - 97.7^\circ) \text{ A}$$

$$P_{Lu} = 0 \text{ W}$$

$$Q_{Lu} = \omega L_u |\underline{I}_{Lu}|^2 = 0.67 \text{ mVAR}$$